

The empirical recurrence for OEIS A301825, recovered from its tail

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Abstract

OEIS A301825 records “Empirical recurrence of order 66” and keeps the recurrence itself in a linked file rather than in the entry, so the conjecture cannot be read off the entry at all. It can still be settled. The entry counts arrays under a condition local to a bounded window of lines, which makes the count a walk count on a finite digraph and forces a monic integer recurrence of order at most the number $S' = 112$ of states with distinct futures. Berlekamp–Massey on enough exact terms therefore returns the minimal such recurrence, and on the tail beginning at $n = 7$, so the recurrence is proved for every $n \geq 73$. Its last coefficient is $c_{66} = -36 \neq 0$, so no further tail satisfies anything shorter; any order-66 recurrence the sequence satisfies from any point on must then have the same characteristic polynomial. The entry’s recurrence is the one displayed here, whatever its linked file contains, and it is proved.

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1 A conjecture that is not written down

OEIS A301825 is “Number of $4 \times n$ 0..1 arrays with every element equal to 1 or 2 horizontally or antidiagonally adjacent elements, with upper left element zero.”. It has offset 1 and begins

0, 13, 8, 68, 299, 1351, 6653, 30441, ...

The entry states:

Empirical recurrence of order 66 (see link above)

As of the “Last modified” line on the live entry (Mar 27 2018 10:15:37, revision 4) this is still recorded as empirical, and it is still open. The coefficients are nowhere in the entry text: what is conjectured is only that a recurrence of that order exists.

2 The count is a walk count

The entry’s condition is local. It constrains a bounded window of consecutive lines of the array and nothing beyond it, so the admissible configurations of one window are the vertices of a finite digraph, an edge joins u to v when v may follow u , and an admissible array is exactly a walk. Writing M for the adjacency matrix and ι, τ for the indicator vectors of the admissible first and last windows,

$$a(n) = \iota^\top M^{j(n)} \tau$$

for a linear reindexing j fixed by the entry’s shape. States with identical futures contribute identically to that product and may be merged without changing any count; after merging $S' = 112$ states remain, and it is that bound every computation below uses.

Lemma 1. *a satisfies a monic linear recurrence with integer coefficients of order at most $S' = 112$.*

Proof. By the Cayley–Hamilton theorem $M^{S'}$ is an integer combination of $I, M, \dots, M^{S'-1}$, and multiplying that relation on the left by $\iota^\top M^j$ and on the right by τ turns it into a relation among $S' + 1$ consecutive values of a . $\square$

3 Why the question has to be asked of a tail

Running Berlekamp–Massey on the sequence from its first term returns an order LARGER than 66: the opening terms do not satisfy the recurrence, and a recurrence that holds only eventually always looks longer when the exceptional terms are included. That is not evidence against the entry. An OEIS empirical recurrence is asserted for n beyond some index, not from the first term, so the question has to be asked of the tails. It was asked of each tail in turn, and the tail beginning at $n = 7$ is the first whose minimal order is exactly the stated 66.

Lemma 2. *A sequence known to satisfy some linear recurrence of order at most S' is determined, together with its minimal recurrence, by any $2S'$ consecutive terms: Berlekamp–Massey applied to them returns that minimal recurrence exactly.*

Applied to $2S' = 224$ exact terms from $n = 7$ on, it returns order 66 — the order the entry states — with integer coefficients:

$$a(n) = \sum_{i=1}^{66} c_i a(n-i),$$

c_1	4	c_{18}	288501	c_{35}	−1096988	c_{52}	−4159571
c_2	17	c_{19}	1054541	c_{36}	−53065881	c_{53}	2683946
c_3	−47	c_{20}	−1067891	c_{37}	69697302	c_{54}	705425
c_4	−159	c_{21}	−14188188	c_{38}	12464537	c_{55}	−1595187
c_5	202	c_{22}	−6063103	c_{39}	−34519473	c_{56}	747501
c_6	1023	c_{23}	24915776	c_{40}	2216297	c_{57}	−571304
c_7	−330	c_{24}	28699105	c_{41}	−14383614	c_{58}	128213
c_8	−4037	c_{25}	−18047891	c_{42}	28687841	c_{59}	241076
c_9	−1629	c_{26}	15163815	c_{43}	−23669350	c_{60}	−102249
c_{10}	7461	c_{27}	−31279494	c_{44}	10451123	c_{61}	4855
c_{11}	4870	c_{28}	−49599170	c_{45}	−6290025	c_{62}	−3457
c_{12}	16502	c_{29}	85744164	c_{46}	3034617	c_{63}	−2592
c_{13}	55776	c_{30}	−83011082	c_{47}	−415970	c_{64}	2496
c_{14}	−143416	c_{31}	73657898	c_{48}	−2255292	c_{65}	−252
c_{15}	−351002	c_{32}	−114436868	c_{49}	5246838	c_{66}	−36
c_{16}	234895	c_{33}	126312618	c_{50}	−2626836		
c_{17}	938606	c_{34}	−32856752	c_{51}	2401296		

4 The recurrence, and the identification

Theorem 3. *The recurrence above holds for every $n \geq 73$, and it is the recurrence OEIS A301825 records.*

Proof. That it holds is Lemma 2 applied to the tail from $n = 7$, whose minimal recurrence it is.

For the identification, write $q_{\min}$ for the characteristic polynomial of that minimal recurrence, $x^{66} - c_1 x^{65} - \dots - c_{66}$. Its constant term is $-c_{66} = 36$, which is not zero, so $q_{\min}(0) \neq 0$ and $q_{\min}$ has no root at the origin. A solution of a constant-coefficient recurrence is a combination of terms $n^k \lambda^n$ with λ a root of its characteristic polynomial, and only the components with $\lambda = 0$

vanish after finitely many terms. Since $q_{\min}$ has none, no component of a dies out, and the minimal recurrence of every further tail is again $q_{\min}$ — the order cannot drop below 66 at any later starting point.

Now let q be the characteristic polynomial of whatever recurrence the entry asserts. The entry asserts it has order 66, so $\deg q = 66$, and it is monic. It holds from some index t on. From $\max(t, 7)$ on, the sequence satisfies both q and $q_{\min}$, and the minimal polynomial of that tail is $q_{\min}$ by the previous paragraph; therefore $q_{\min} \mid q$. Both are monic of degree 66, so $q = q_{\min}$. $\square$

5 Verification

Four checks, all in exact integer or rational arithmetic.

First, the walk model reproduces all 23 terms the entry publishes, exactly. This is what ties the model to the entry, and it is the only place where reading the entry’s English enters the argument.

Second, the order was computed twice by independent routes: once modulo a 61-bit prime, where every intermediate is a machine word, and once over $\mathbb{Q}$ in exact rational arithmetic. Both return 66, and the exact run additionally confirms every coefficient is an integer — had any been a proper fraction the recurrence would not be the entry’s.

Third, the recovered recurrence was evaluated directly on the entry’s own published terms, with no matrices and no Berlekamp–Massey involved. It holds at each of the 0 indices where the entry publishes enough earlier terms to test it.

Fourth, the last coefficient was checked to be nonzero, which is what the identification above turns on; it is -36 .

Remark 4. The conjecture’s own statement was never read. The entry pins it down to the family of order-66 recurrences this sequence satisfies from some point on, and that family turns out to have exactly one member.

References

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